Registers

A/PSW	0x FF A5
BC	0x 00 04
DE	0x 00 04
HL	0x 00 04
SP	0x 47 FF
PC	0x 00 4E

Flags

S	☑
Z	☐
AC	☐
P	☑
C	☑
V	☐

```
1  LXI SP,47FFH
2  LHLD 4530H
3  XCHG
4  LHLD 4532H
5  MOV B,H
6  MOV C,L
7  LXI H,0000H
8  SHLD 4536H
9  MOV A,B
10 ORA C
11 JZ STORE
12 MLOOP: DAD D
13 JNC NOCY
14 PUSH H
15 LHLD 4536H
16 INX H
17 SHLD 4536H
18 POP H
19 NOCY: DCX B
20 MOV A,B
21 ORA C
22 JNZ MLOOP
23 STORE: SHLD 4534H
24 LHLD 453AH
25 XCHG
26 LHLD 4538H
27 LXI B,0000H
28 DLOOP: MOV A,L
29 SUB E
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	2A 30 45	LHLD 4530H
3	0x6	EB	XCHG
4	0x7	2A 32 45	LHLD 4532H
5	0xA	44	MOV B,H
6	0xB	4D	MOV C,L
7	0xC	21 00 00	LXI H,0000H
8	0xF	22 36 45	SHLD 4536H
9	0x12	78	MOV A,B
10	0x13	B1	ORA C
11	0x14	CA 2A 00	JZ STORE
12	0x17	19	MLOOP: DAD D
13	0x18	D2 24 00	JNC NOCY
14	0x1B	E5	PUSH H
15	0x1C	2A 36 45	LHLD 4536H
16	0x1F	23	INX H

Displaying Memory Locations from 0x4530 to 0x453F

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4530	05
0x4531	00
0x4532	03
0x4533	00
0x4534	0F
0x4535	00
0x4536	00
0x4537	00
0x4538	11
0x4539	00
0x453A	04
0x453B	00
0x453C	04
0x453D	00

```
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16 INX H
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(1) Multiplication (x x y)

Given:

X = 0005H = 5

Y = 0003H = 3

Calculation:

X x Y = 5 x 3 = 15

15_{10} = 000FH

Product = 0000000FH

(2) Division

Given:

Dividend = 0011H = 17

Divisor = 0004H = 4

17 ÷ 4 = 4 remainder 1 (or)

17 = (4 x 4) + 1

= 16 + 1

Quotient = 0004H

Remainder = 0001H